HARNESS ARCHITECTURE DOCUMENT

Ops Runbook Harness

An AI agent harness that diagnoses and remediates infrastructure incidents using pre-approved runbooks — with safety constraints the agent cannot override.

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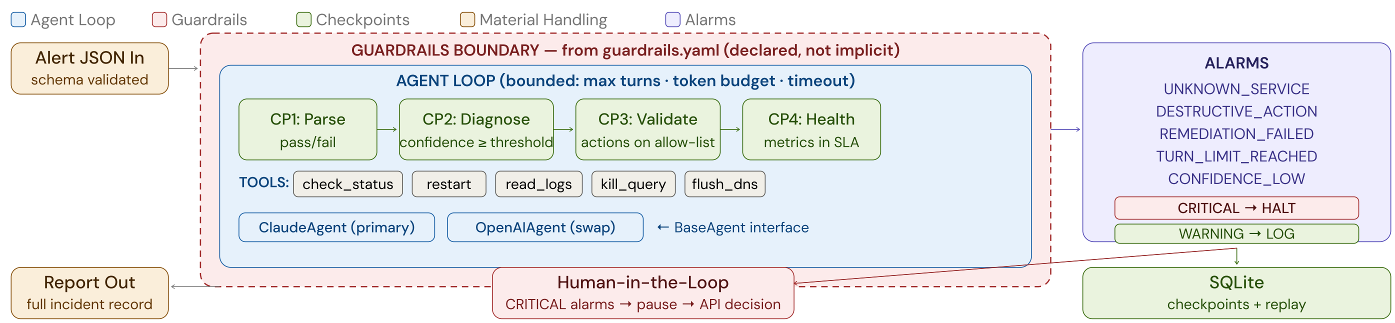
Stack: Python · FastAPI · Claude / OpenAI · SQLite · Pydantic

1. OVERVIEW

When production systems break, on-call engineers get paged for problems with known fixes. The hard part isn't running the fix — it's knowing whether automation can **safely act**, verifying the fix **actually worked**, and catching **downstream damage** before it cascades.

This harness governs an AI agent that follows pre-approved runbooks to diagnose and remediate incidents. The agent proposes actions; the **harness decides what's allowed**. Four architecturally separate pillars — each a distinct module — enforce safety, evaluate quality, structure data, and signal failures.



End-to-end architecture. Guardrails wrap the loop. Checkpoints gate each stage. Alarms halt on critical. All four pillars are separate modules.

2. THE FOUR PILLARS

PILLAR	MODULE	KEY CONTRACT
Guardrails	`harness/guardrails.py`	Returns `Allowed`, `Blocked`, or `NeedsApproval` for every action. Loads YAML config; enforces turn limits, token budgets, timeouts.
Checkpoints	`harness/checkpoints.py`	Each CP returns `passed: bool` + state snapshot to SQLite. Replay loads prior state and skips completed stages.
Material Handling	`harness/material.py`	Validates incoming alert JSON (Pydantic). Invalid input rejected before loop starts. Output is a complete incident record.
Alarms	`harness/alarms.py`	Each alarm: `{type, context, severity, action}`. Five named types. CRITICAL = halt + escalate. WARNING = log.

3. DESIGN PRINCIPLES

Pillar Separation

Each pillar is a distinct Python module. The agent never touches guardrail config, checkpoint persistence, or alarm emission directly.

Declared, Not Implicit

Guardrails live in `guardrails.yaml`, not prompt engineering. Change what's allowed without changing code.

Block First, Ask Second

Any action not on the allow-list is blocked before execution. The AI can only do what's pre-approved.

Swappable Agent

`BaseAgent` abstract interface. Claude or OpenAI — zero harness changes. The harness is the product, not the model.

4. KEY DECISIONS

Adaptive feedback loop: When the guardrail blocks an action, the agent receives a structured "blocked" result as its next message — not an error. The agent reasons about the rejection and chooses a different approach. Harness constrains; agent adapts.

Checkpoint replay: Every checkpoint persists full state to SQLite. To replay from CP3, the harness loads CP2's saved state and skips stages 1–2. Re-run remediation without re-diagnosing — critical for debugging.

The core safety contract

The agent proposes. The harness disposes. No action executes without passing the guardrail allow-list. No checkpoint passes without explicit criteria. No critical failure goes unescalated. The model is the engine — the harness is the car.